

Recitation: Partial Fractions Solutions

Problem 1. Find the partial fractions decomposition for the following:

(a) $\frac{x+13}{x^2-4x-5}$

$$\begin{aligned}\frac{x+13}{x^2-4x-5} &= \frac{x+13}{(x-5)(x+1)} = \frac{A}{x-5} + \frac{B}{x+1} \\ &= \frac{A(x+1) + B(x-5)}{(x-5)(x+1)} \\ &= \frac{Ax + A + Bx - 5B}{(x-5)(x+1)}\end{aligned}$$

But then $x+13 = Ax + A + Bx - 5B$. So, we have...

$$\begin{aligned}x: \quad A + B &= 1 \\ 1: \quad A - 5B &= 13\end{aligned}$$

Subtracting these equations, we have $6B = -12$, so $B = -2$. Using this in the first equation, we have $A - 2 = 1$, i.e. $A = 3$. Therefore,...

$$\frac{x+13}{x^2-4x-5} = \boxed{\frac{3}{x-5} + \frac{-2}{x+1}}$$

We can also find A, B with Heaviside's: $A = \frac{5+13}{5+1} = \frac{18}{6} = 3$ and $B = \frac{-1+13}{-1-5} = \frac{12}{-6} = -2$.

(b) $\frac{3x+13}{2x^2+4x-6}$

$$\begin{aligned}\frac{3x+13}{2x^2+4x-6} &= \frac{3x+13}{2(x-1)(x+3)} = \frac{1}{2} \cdot \frac{3x+13}{(x-1)(x+3)} = \frac{1}{2} \left[\frac{A}{x-1} + \frac{B}{x+3} \right] \\ &= \frac{1}{2} \left[\frac{A(x+3) + B(x-1)}{(x-1)(x+3)} \right] \\ &= \frac{1}{2} \left[\frac{Ax + 3A + Bx - B}{(x-1)(x+3)} \right]\end{aligned}$$

But then $3x+13 = Ax + 3A + Bx - B$. So, we have...

$$\begin{aligned}x: \quad A + B &= 3 \\ 1: \quad 3A - B &= 13\end{aligned}$$

Add these equations, we have $4A = 16$, so $A = 4$. Using this in the first equation, we have $4 + B = 3$, i.e. $B = -1$. Therefore,...

$$\frac{3x+13}{2x^2+4x-6} = \boxed{\frac{1}{2} \left[\frac{4}{x-1} + \frac{-1}{x+3} \right]}$$

We can also find A, B with Heaviside's: $A = \frac{3(1)+13}{1+3} = \frac{16}{4} = 4$ and $B = \frac{3(-3)+13}{-3-1} = \frac{4}{-4} = -1$.

(c) $\frac{5x^2 - 3x + 4}{(x-1)(x^2+1)}$

$$\begin{aligned}\frac{5x^2 - 3x + 4}{(x-1)(x^2+1)} &= \frac{A}{x-1} + \frac{Bx+C}{x^2+1} \\ &= \frac{A(x^2+1) + (Bx+C)(x-1)}{(x-1)(x^2+1)} \\ &= \frac{(Ax^2 + A) + (Bx^2 - Bx + Cx - C)}{(x-1)(x^2+1)}\end{aligned}$$

But then $5x^2 - 3x + 4 = Ax^2 + A + Bx^2 - Bx + Cx - C$. So, we have...

$$\begin{aligned}x^2: \quad & A + B = 5 \\ x: \quad & -B + C = -3 \\ 1: \quad & A - C = 4\end{aligned}$$

Subtracting the first and third equation, we have $B+C = 1$. But then $B+C = 1$ and $-B+C = -3$ (the middle equation). Adding these equations, we find $2C = -2$, i.e. $C = -1$. But then because $B+C = 1$, we know $B-1 = 1$, i.e. $B = 2$. But because $A+B = 5$, we know $A+2 = 5$, i.e. $A = 3$. Therefore, we have...

$$\boxed{\frac{5x^2 - 3x + 4}{(x-1)(x^2+1)} = \frac{3}{x-1} + \frac{2x-1}{x^2+1}}$$

We can also find A with Heaviside's: $A = \frac{5-3+4}{1+1} = \frac{6}{2} = 3$.

Problem 2. Find the partial fraction decomposition of the following:

$$\frac{3x^4 - x^3 + 7x^2 - 4x + 2}{x^3 + 2x}$$

Solution. Observe that the degree of the numerator—4—is not smaller than the degree of the denominator—3. So, we first need to polynomial long divide. We have...

$$\begin{array}{r} 3x - 1 \\ x^3 + 2x \overline{) 3x^4 - x^3 + 7x^2 - 4x + 2} \\ \underline{- 3x^4} - 6x^2 \\ - x^3 + x^2 - 4x \\ \underline{x^3} + 2x \\ x^2 - 2x + 2 \end{array}$$

Therefore, we have quotient $3x - 1$ and remainder $x^2 - 2x + 2$. So, we have $\frac{3x^4 - x^3 + 7x^2 - 4x + 2}{x^3 + 2x} = 3x - 1 + \frac{x^2 - 2x + 2}{x^3 + 2x}$. We need to find the partial fraction decomposition of $\frac{x^2 - 2x + 2}{x^3 + 2x}$. The denominator factors as $x(x^2 + 2)$. We have...

$$\begin{aligned} \frac{x^2 - 2x + 2}{x(x^2 + 2)} &= \frac{A}{x} + \frac{Bx + C}{x^2 + 2} \\ &= \frac{A(x^2 + 2) + (Bx + C)x}{x(x^2 + 2)} \\ &= \frac{Ax^2 + 2A + Bx^2 + Cx}{x(x^2 + 2)} \end{aligned}$$

Therefore, we must have $x^2 - 2x + 2 = Ax^2 + 2A + Bx^2 + Cx$. Relating coefficients, we must have...

$$\begin{aligned} x^2: \quad & A + B = 1 \\ x: \quad & C = -2 \\ 1: \quad & 2A = 2 \end{aligned}$$

From the last equation, we see $A = 1$. [We can also see this with Heaviside's: $A = \frac{0-0+2}{0^2+2} = \frac{2}{2} = 1$.] Using this in the first equation, we have $1 + B = 1$ so $B = 0$. Therefore, we have...

$$\boxed{\frac{3x^4 - x^3 + 7x^2 - 4x + 2}{x^3 + 2x} = 3x - 1 + \frac{-1}{x} + \frac{-2}{x^2 + 2}}$$

Problem 3. Complete the following:

- (a) Find the partial fractions decomposition of $\frac{x^2 - x + 2}{x^3}$ ‘immediately.’
- (b) Find a rational function with the partial fraction decomposition $2 + \frac{-3}{x+1} + \frac{1}{x-2}$.

Solution.

- (a) We know the decomposition has the form $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3}$ for some A, B, C . But this is precisely the form we have when we divide x^3 into the numerator! Doing so, we have...

$$\frac{x^2 - x + 2}{x^3} = \frac{x^2}{x^3} - \frac{x}{x^3} + \frac{2}{x^3} = \frac{1}{x} + \frac{-1}{x^2} + \frac{2}{x^3}$$

So, this is the partial fraction decomposition, i.e. $A = 1$, $B = -1$, and $C = 2$.

- (b) We know that if the given expression was the partial fraction decomposition, putting the rational functions ‘back together’, we would obtain a rational function with that partial fraction decomposition! So, we have...

$$2 + \frac{-3}{x+1} + \frac{1}{x-2} = \frac{2(x+1)(x-2) - 3(x-2) + 1(x+1)}{(x+1)(x-2)} = \frac{2x^2 - 4x + 3}{x^2 - x - 2}$$

Therefore, $\frac{2x^2 - 4x + 3}{x^2 - x - 2}$ has the given partial fraction decomposition.